WakeBand rigid rev A: 4-layer 0.8 mm FR4 board, about 15 x 33 mm, parts side down over a 60 mAh cell and an 8 mm LRA.
Cell: 60 mAh protected LiPo, about 16 x 13 x 4-5 mm, wire leads (X2 Robotics; alternate Ampul 401215). Pod about 17.5 x 35 x 9 mm.
Sheets connect through global labels and power nets. Same circuit as flex rev A; only the cell, the dock pad footprint and the layout changed.

power

Power: nPM1304 PMIC, battery, NTC, ship mode, current jumper

File: power.kicad_sch

mcu

MCU: ISP2454 (nRF54L15) module, SWD, UART, I2C, interrupts

File: mcu.kicad_sch

sensors

Sensors: LIS2DW12 accelerometer

File: sensors.kicad_sch

haptics

Haptics: DRV2605L driver and LRA pads

File: haptics.kicad_sch

dock

Dock: six pogo pads, TVS, series resistors

File: dock.kicad_sch

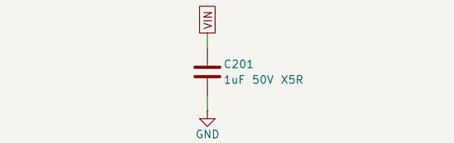
debug

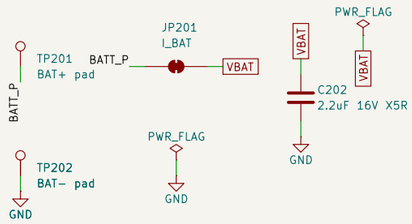
Debug: test pads, debug LED (DNP option)

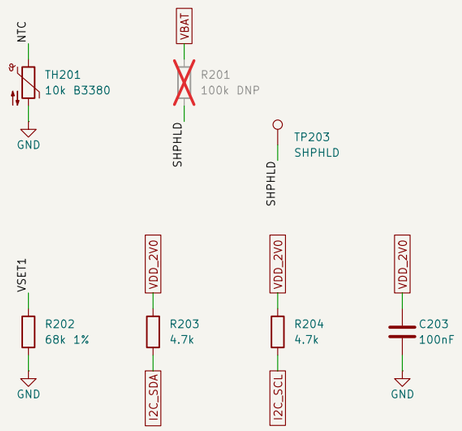
File: debug.kicad_sch

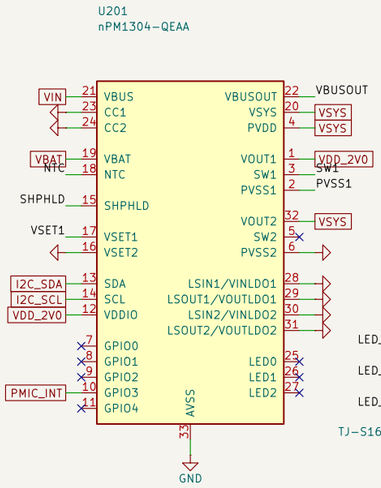
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /
File: rigid-rev-a.kicad_sch
Title: WakeBand rigid rev A - Top
Size: A4 | Date: 2026-11-02 | **Rev: RA0 (rigid 4-layer, 60**
KiCad E.D.A. 10.0.3 | Id: 1/7

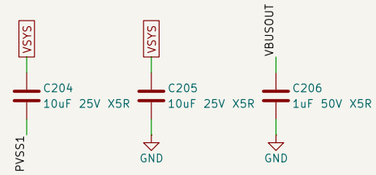
nPM1304 power sheet (step 3). Based on Nordic nPM1304 datasheet, reference circuit 'Configuration 2' (QFN):
one buck, no LDO/load switch, ship-mode exit and NTC used. Draft until the Stage 0b nPM1304-EK tests.

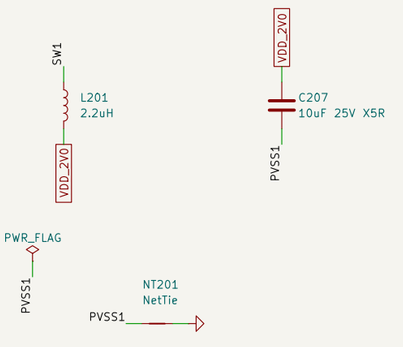












Strapping and connections (from the datasheet):
- VBUS from dock pad VIN. VBUS abs max 22 V; OVP 5.5 V typ; VBUSPOR 3.9 V. The VIN TVS (D601) is on the dock sheet, at the pad.
- CC1/CC2 to GND: no USB-C; VBUS limit stays at the 100 mA default (enough for the 30 mA charge current).
- VBUSOUT not used by the SoC but needs its 1 uF decoupling cap.
- VSET1 = 68k -> BUCK1 starts at 2.0 V (datasheet table: 68k = 2.0 V, max 5% tolerance). VOUT1 = VDD_2V0.
  2.0 V (not 1.8 V) leaves margin over the ISP2454 1.7 V minimum when the buck is 5 % low (1.9 V).
- BUCK2 unused: VSET2 to GND (off), VOUT2 to VSYS, SW2 open, PVSS2 to GND, no inductor (as in reference config 2).
- LOADSW1/2 unused: LSIN1, LSOUT1, LSIN2, LSOUT2 to GND (as in reference config 3). LED0-2 open.
- VDDIO from VOUT1. Unused GPIO0/1/2/4 open: default is input with weak pull-down.
- PMIC_INT on GPIO3 (any GPIO can be the interrupt; set in devicetree).
- PVSS1 joins GND through net tie NT201 (one via to the GND layer), as in the reference layout.
- NTC: 10k 1% B25/50 = 3380 K, one of the three supported types; JEITA defaults match it. Max 100 pF across it.
- Ship mode exits on SHPHLD low (button to GND) OR VBUS > VBUSPOR, so docking alone wakes the band
  (still confirm on the EK in Stage 0b).
- DECIDED 2026-10-30: SHPHLD relies on the nPM1304's internal 50k pull-up. R201 (extra pull-up to VBAT) and
  SW201 (button) stay DNP; TP203 is the way to pull SHPHLD low on the bench.
- TWI address 0x6B. I2C pull-ups R203/R204 serve the whole 2.0 V bus (4.7k for 400 kHz).
- JP201: cut to put the PPK2 in series with the cell; bridge again with solder.
- CELL (rigid rev A): 60 mAh protected LiPo, ~16 x 13 x 4-5 mm, wire leads to TP201 (+) / TP202 (-).
  Source X2 Robotics (x2robotics.ca, 3.7 V 60 mAh); alternate Ampul 401215 (4 x 12 x 15 mm).
  Charge at <= 30 mA (0.5C). nPM1304 power-on defaults: charging disabled, ICHG 4 mA, VTERM 4.0 V.
  Firmware must set ICHG = 30 mA and VTERM = 4.2 V over I2C, then enable charging.

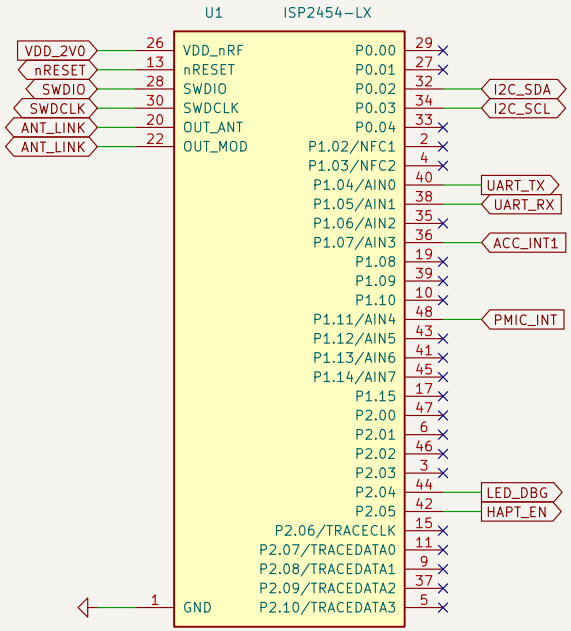
(check) before freezing:
- U201 footprint: KiCad QFN-32 5x5 EP 3.45 mm; datasheet EP is 3.5 nom (3.4-3.6). Compare with Nordic's QFN reference layout.
- U201 has no LCSC stock (only the nPM1304-EK, C50090302); source the chip elsewhere or as a consigned part.
- L201: Murata DFE201610E-2R2M confirmed, on KiCad's L_Murata_DFE201610P land pattern (not yet compared with Murata's drawing).
- GPIO3 for PMIC_INT: match the nPM1304-EK / Zephyr devicetree used in Stage 0b.
- VSYS abs max 5.5 V: dock supply must stay at 5.0 V; DRV2605L blocked while VBUS is present.

nPM1304 per Nordic reference circuit configuration 2 (QFN32)
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /power/
File: power.kicad_sch

Title: WakeBand rigid rev A — Power

Size: A3	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 6C**
KiCad E.D.A. 10.0.3		Id: 2/7

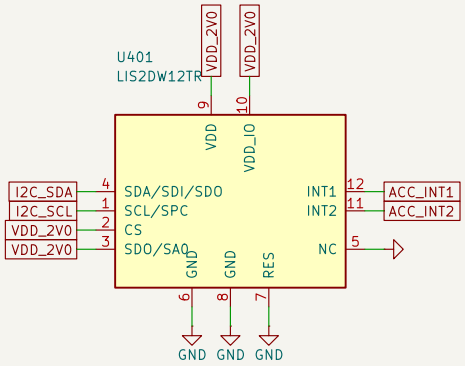
ISP2454 notes (datasheet R7):
- Antenna, 32 MHz + 32.768 kHz crystals, DC-DC inductor and all decoupling are inside. No external parts.
- OUT_ANT (20) must be joined to OUT_MOD (22) by a short trace (net ANT_LINK).
- Pads 49-62 are mechanical: isolated copper in the footprint, not in this symbol.
- Antenna corner 1 mm from the board edge, no copper or parts on any layer in the keep-out,
  and keep the cell and LRA 2 mm (plan view) clear of it.
- Firmware: set LFXO load capacitance to 19 pF (devicetree load-capacitance-femtofarad = <19000>).
- Signals use OUTER-row pads. Rigid rev A: nRESET (13) and the trapped GND pads (12, 14, 16, 18, 21, 23)
  use epoxy-filled, copper-capped vias in pad (JLCPCB POFV). Pad 10 stays NC.
- I2C on TWIM30 (P0, low-power domain). UART on UARTE20 (P1).
- P1.02/P1.03 are NFC pins: left unused. P2 trace pins left unused.
- GPIO plan and reasons: docs/flex-rev-a-pinout.md

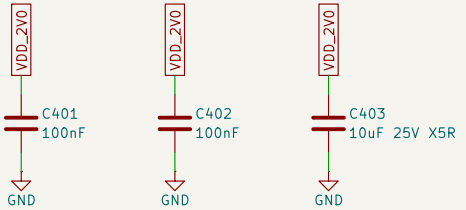


Review 2026-10-30:
- Pad 10 (P1.10) is not connected: it sits in the lower outer row, which cannot be routed without via-in-pad
  (antenna keep-out below, 0.25 mm gaps to the inner pads). The DRV2605L IN/TRIG is tied to GND; effects run over I2C.
- GND pads 12, 14, 16 and 18 sit in the trapped lower row. They are tied together inside the module. Rigid rev A
  ties each one to the L2 GND plane with a filled, capped via in pad (POFV). Pads 8 and 24 reach the large pads 7 and 25 by a short trace.
  All of them stay on the GND net in the schematic.
- VDD_nRF is VDD_2V0 (BUCK1 at 2.0 V, margin over the 1.7 V minimum when the buck is 5 % low).

Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /mcu/
File: mcu.kicad_sch
Title: WakeBand rigid rev A — MCU
Size: A4 | Date: 2026-11-02 | **Rev: RA0 (rigid 4-layer, 60**
KiCad E.D.A. 10.0.3 | Id: 3/7

Sensors sheet (step 5): ST LIS2DW12 on the shared 2.0 V I2C bus. Pinout from ST datasheet DocID029682 Rev 4, table 2.







- VDD (pin 9) and VDD_IO (pin 10) on VDD_2V0: C401 at pin 9 (VDD), C402 at pin 10 (VDD_IO).
- C403 10 uF bulk on VDD near pin 9, per ST's application hint (100 nF + 10 uF).
- CS (pin 2) tied to VDD_2V0: I2C mode (SPI disabled).
- SDO/SA0 (pin 3) tied to VDD_2V0: I2C address 0011001b = 0x19 (0x18 if SA0 were low).
  CS and SA0 also have internal pull-ups (about 49k at 1.8 V, 30k at 2.5 V); the hard strap avoids depending on them.
- SDA/SCL on I2C_SDA/I2C_SCL. No pull-ups here: R203/R204 on the power sheet serve the whole bus.
- INT1 -> ACC_INT1 (SoC wake GPIO P1.07). INT2 -> ACC_INT2, test pad TP401 only.
- Pin 5 NC: datasheet allows VDD, VDD_IO or GND; tied to GND. Pin 7 RES: connect to GND (datasheet).
- Pins 6 and 8 GND.

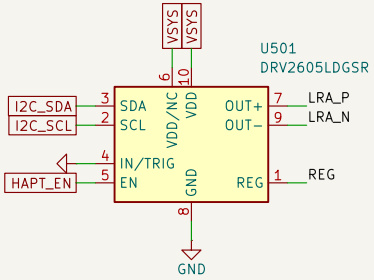
LIS2DW12 per ST datasheet DocID029682 Rev 4
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

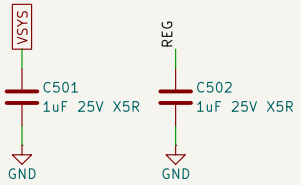
Sheet: /sensors/
File: sensors.kicad_sch

Title: WakeBand rigid rev A — Sensors

Size: A4	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60**
KiCad E.D.A. 10.0.3		Id: 4/7

Haptics sheet (step 5): TI DRV2605L (VSSOP-10, DGS) on VSYS, I2C at 2.0 V. From TI datasheet SLOS854D.







- VDD (pin 10) from VSYS with C501 1 uF; pin 6 VDD/NC tied to VDD as the datasheet allows.
- REG (pin 1): internal 1.8 V regulator output, C502 1 uF required.
- EN (pin 5) <- HAPT_EN (P2.05).
- IN/TRIG (pin 4) tied to GND (datasheet: connect to GND if not used). Effects are triggered over I2C (GO bit). HAPT_TRIG was dropped: ISP2454 pad 10 (P1.10) is in the lower outer row and cannot be routed without via-in-pad.
- SDA/SCL on the shared bus; VIH = 1.3 V, so 2.0 V logic is fine. I2C address 0x5A.
- OUT+ (7) / OUT- (9) to the LRA lead pads TP501 (LRA+) and TP502 (LRA-), 1.0 x 1.5 mm, hand-soldered.
- MECHANICAL (rigid rev A): the 8 mm LRA is glued to the case floor under the board, not a schematic part. Motor: Vybronics VG0832012. Leads hand-soldered to TP501/TP502 on F.Cu.
- DO NOT switch the driver's VDD off: SDA, SCL, EN and IN/TRIG are limited to VDD + 0.3 V, so an unpowered DRV2605L would be back-powered through its I2C pins and clamp the shared bus. Rev A keeps VDD on with EN low (about 4 uA). Firmware blocks the motor while VBUS is present (VSYS then follows VBUS up to 5.5 V, above the 5.2 V recommended maximum; abs max is 5.5 V).

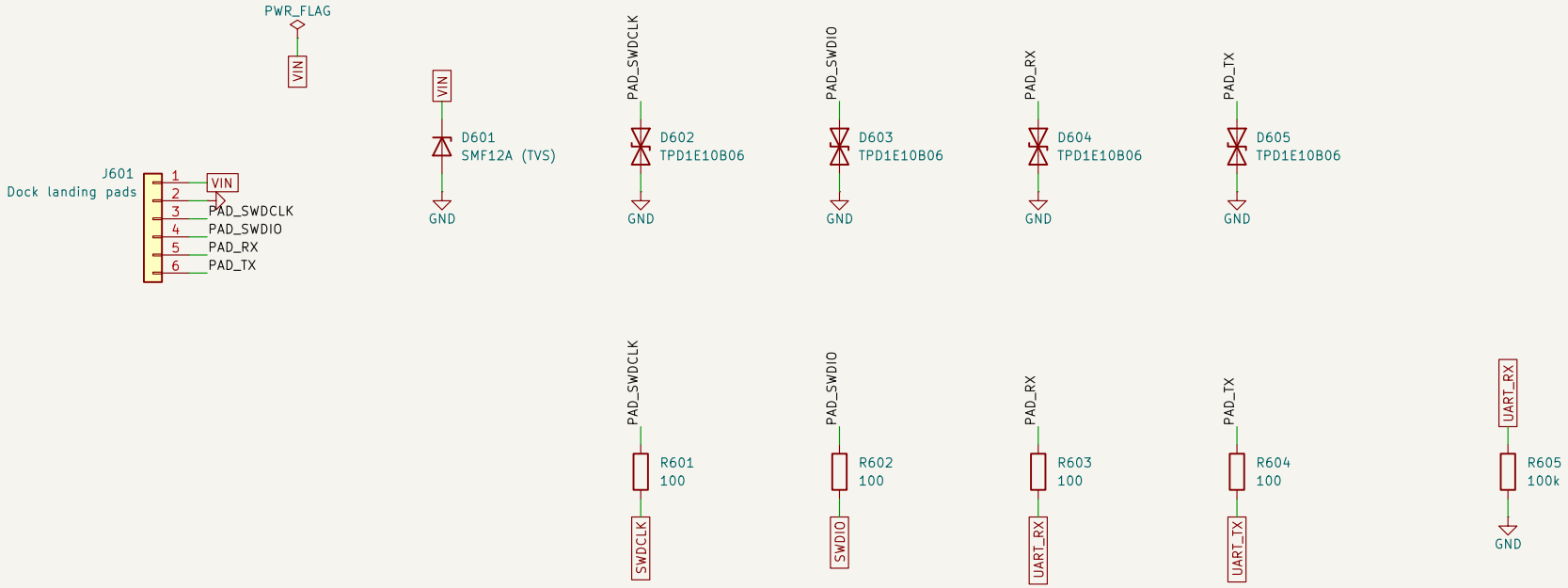
DRV2605L per TI datasheet SLOS854D
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

Sheet: /haptics/
File: haptics.kicad_sch

Title: WakeBand rigid rev A — Haptics

Size: A4	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60**
KiCad E.D.A. 10.0.3		Id: 5/7

Dock sheet (step 6): six round 1.6 mm pogo pads on B.Cu (the pod outer face), 2.5 mm pitch (J601 footprint, rigid rev A).



Pad order: 1 VIN, 2 GND, 3 SWDCLK, 4 SWDIO, 5 UART_RX, 6 UART_TX.
- TVS at the pad on VIN (D601) and on each data pad (D602–D605). No TVS on pad 2: it is the GND reference.
- D601 is Littelfuse SMF12A (SOD-123FL, unidirectional, 12 V standoff, 13.3–14.7 V breakdown at 1 mA, 2.5 uA max at 12 V), clamping at 19.9 V max at 10.1 A: below the nPM1304 VBUS abs max of 22 V, and a wrong 12 V supply does not burn it. The nPM1304 cuts VSYS off above 5.5 V.
  D601 replaces the VIN TVS that was on the power sheet (the pads and the PMIC are on different islands).
- 100 ohm in series between each data pad and the SoC net (R601–R604), TVS on the pad side.
- R605 100k pull-down on UART_RX at the SoC side, so the RX input does not float off the dock.
- Firmware enables the UART only while the PMIC reports VBUS present.
- SWDIO has an always-on internal pull-up in the nRF54L15, so pad 4 sits at 2.0 V on the skin all night.
- Pads: ENIG for rev A (JLCPCB rigid); hard gold (PCBWay) if pogo wear shows.

Pad order from docs/hardware-roadmap.md
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

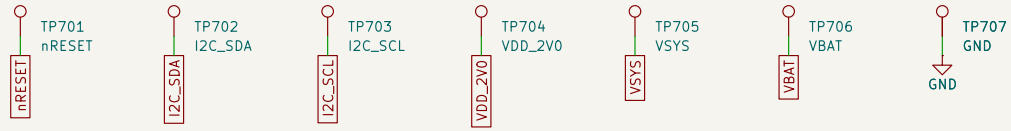
Sheet: /dock/
File: dock.kicad_sch

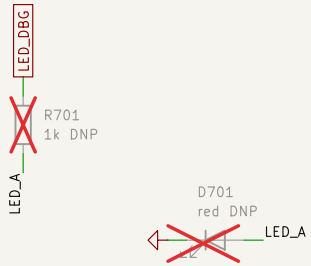
Title: WakeBand rigid rev A – Dock

Size: A4 | Date: 2026-11-02 | **Rev: RA0 (rigid 4-layer, 60**
KiCad E.D.A. 10.0.3 | Id: 6/7

Notes:

Debug sheet (step 6): 0.8 mm round test pads (no paste) on F.Cu, and a DNP debug LED.





- Test pads: nRESET, I2C_SDA, I2C_SCL, VDD_2V0, VSYS, VBAT, GND (TP701–TP707).
  SHPHLD (TP203, power sheet) and ACC_INT2 (TP401, sensors sheet) already have pads.
- nRESET: test pad only. The nRF54L15 has an internal pull-up on nRESET, and SWD can reset the chip,
  so nothing on the board drives this input. ERC: the passive test pad satisfies KiCad's 'input not driven'
  check, so the earlier nRESET error is gone without any exclusion. This is intended, not a missing driver.
- LED_DBG (P2.04) -> R701 1k -> D701 (0402 red) -> GND, both DNP. Fit them only for bench bring-up:
  a lit LED would cost far more than the sleep budget.

Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /debug/
File: debug.kicad_sch
Title: WakeBand rigid rev A – Debug
Size: A4 | Date: 2026-11-02 | **Rev: RA0 (rigid 4-layer, 60**
KiCad E.D.A. 10.0.3 | Id: 7/7